

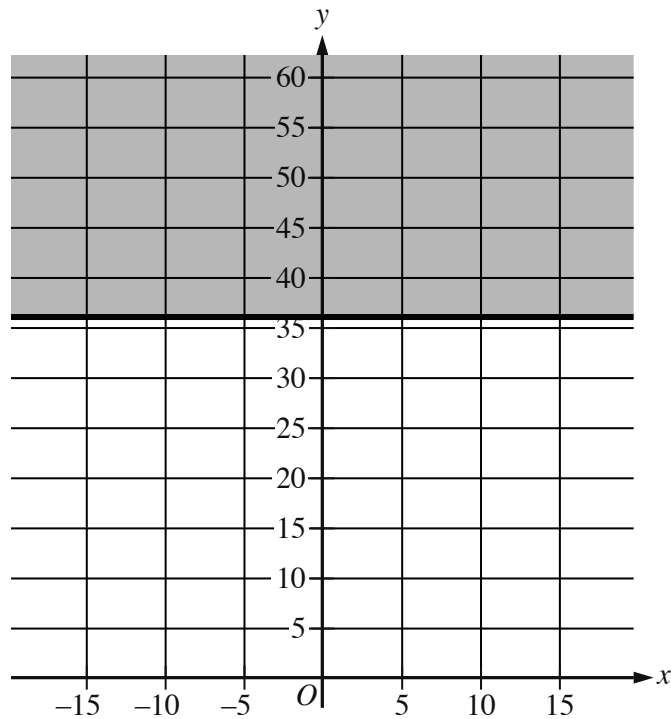
● EASY

● ALGEBRA

Linear inequalities in one or two variables

02

10 questions · ~15 min



- The boundary of the inequality is a solid line.
 - The line is horizontal.
 - The line passes through the following points:
 - (negative 15 comma 36)
 - (0 comma 36)
 - (15 comma 36)
- The area above the boundary is shaded.

The shaded region shown in the graph represents all the solutions to which inequality?

- A. $x \leq 36$
- B. $x \geq 36$
- C. $y \leq 36$

D. $y \geq 36$

Q2

ALGEBRA

A cleaning service that cleans both offices and homes can clean at most 14 places per day. Which inequality represents this situation, where f is the number of offices and h is the number of homes?

A. $f + h \leq 14$

B. $f + h \geq 14$

C. $f - h \leq 14$

D. $f - h \geq 14$

Q3

ALGEBRA

During a portion of a flight, a small airplane's cruising speed varied between 150 miles per hour and 170 miles per hour. Which inequality best represents this situation, where s is the cruising speed, in miles per hour, during this portion of the flight?

A. $s \leq 20$

B. $s \leq 150$

C. $s \leq 170$

D. $150 \leq s \leq 170$

Q4

ALGEBRA

A clothing store is having a sale on shirts and pants. During the sale, the cost of each shirt is \$15 and the cost of each pair of pants is \$25. Geoff can spend at most \$120 at the store. If Geoff buys s shirts and p pairs of pants, which of the following must be true?

- A. $15s + 25p \leq 120$
- B. $15s + 25p \geq 120$
- C. $25s + 15p \leq 120$
- D. $25s + 15p \geq 120$

Q5

ALGEBRA

During spring migration, a dragonfly traveled a minimum of 1,510 miles and a maximum of 4,130 miles between stopover locations. Which inequality represents this situation, where d is a possible distance, in miles, this dragonfly traveled between stopover locations during spring migration?

- A. $d \leq 1,510$
- B. $1,510 \leq d \leq 4,130$
- C. $d \geq 4,130$
- D. $4,130 \leq d \leq 5,640$

Q6

ALGEBRA

Which of the following ordered pairs (x, y) satisfies the inequality $5x - 3y < 4$?

1. $(1, 1)$
2. $(2, 5)$
3. $(3, 2)$

- A. I only
- B. II only
- C. I and II only
- D. I and III only

Q7

ALGEBRA

Normal body temperature for an adult is between 97.8°F and 99°F , inclusive. If Kevin, an adult male, has a body temperature that is considered to be normal, which of the following could be his body temperature?

- A. 96.7°F
- B. 97.6°F
- C. 97.9°F
- D. 99.7°F

Monarch butterflies can fly only with a body temperature of at least 55.0 degrees Fahrenheit $^{\circ}\text{F}$. If a monarch butterfly's body temperature is 51.3°F , what is the minimum increase needed in its body temperature, in $^{\circ}\text{F}$, so that it can fly?

- A. 1.3
- B. 3.7
- C. 5.0
- D. 6.3

Maria plans to rent a boat. The boat rental costs \$60 per hour, and she will also have to pay for a water safety course that costs \$10. Maria wants to spend no more than \$280 for the rental and the course. If the boat rental is available only for a whole number of hours, what is the maximum number of hours for which Maria can rent the boat?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The total cost, in dollars, to rent a surfboard consists of a \$ 25 service fee and a \$ 10 per hour rental fee. A person rents a surfboard for t hours and intends to spend a maximum of \$ 75 to rent the surfboard. Which inequality represents this situation?

- A. $10t \leq 75$
- B. $10 + 25t \leq 75$
- C. $25t \leq 75$
- D. $25 + 10t \leq 75$